Assessment of AIS Antenna Placement near Hall-Effect Thrusters for Small Satellite Applications

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Problem Statement

Problem: Dense, partially ionized plumes perturb the local plasma environment, potentially degrading RF payload performance through antenna detuning, phase instabilities, and altered wave propagation—particularly for low-frequency systems like VHF Automatic Identification System (AIS) antennas.

Approach: This study presents a physics-based modelling framework integrating:

- Plume flow-field modelling
- Plasma parameter estimation
- Cold-plasma electromagnetic simulations
- Flux-based surface accumulation models

Key Findings: Low-frequency RF systems (VHF) are significantly more susceptible to plume-induced effects than higher-frequency payloads (S-band, X-band). Contamination from neutralized propellant and charge-exchange ions causes gradual antenna detuning due to dielectric film formation.

Impact: Results underscore the importance of plume-RF coupling analysis in early spacecraft design and motivate mitigation strategies including increased antenna-thruster separation, operational constraints, and protective coatings.

Plume–RF Interaction Challenges

- HET operation introduces **partially ionized xenon plumes** that modify the spacecraft's surrounding plasma environment
- Interaction between ionized plume species and electromagnetic fields becomes a **critical design constraint** for sensitive RF payloads

Frequency Band	Susceptibility	Reason
VHF/UHF (AIS)	**HIGH**	Carrier frequencies near plasma cutoff frequencies
S-band X-band	**LOW**	Wavelengths significantly smaller than plasma perturbation scales

Research Objective

Develop a plume-RF coupling framework to:

- Determine how electric propulsion operation degrades AIS antenna performance
- Propose safe integration guidelines for small satellite missions

Methodology

INTEGRATED MODELLING FRAMEWORK

HALL THRUSTER PLUME MODELLING

- Plume expansion modelling
- Ion/neutral density distribution
- Charge Exchange (CEX) Processes

PLASMA PARAMETER ESTIMATION

- Ion density (n_i)
- Electron Temperature (Te)
- Plasma Frequency (fp)

COLD PLASMA PERMITTIVITY ESTIMATION

- Wave propagation modelling
- Plasma Permittivity Calculation
- Phase Velocity and Attenuation

CONTAMINATION AND DETUNING ANALYSIS

- Flux based xenon accumulation
- Dielectric film formation
- Long term antenna detuning

THRUSTER OPERATING PARAMETERS

- **Power Range:** 40–100 W (typical for small satellites in LEO ~500 km)
- **Propellant:** Xenon
- **Application:** Station-keeping and orbit maintenance

PLUME FLOW FIELD MODELLING

Key Components:

- Quasi-neutral ion-electron dynamics
- Neutral gas expansion
- Charge-exchange (CEX) processes

Spatial Variations Estimated:

- Ion density distribution
- Neutral density distribution
- Ion flux patterns
- Divergence behaviour
- Backflow contamination rates

PLASMA PARAMETER ESTIMATION

Critical Parameters:

1. Plasma Frequency:

$$f_p = \frac{1}{2\pi}\sqrt{\frac{n_e e^2}{\epsilon_0 m_e}}$$

2. Relative Permittivity:

$$\epsilon_r = 1 - \left(\frac{\omega_p}{\omega}\right)^2$$

COLD PLASMA PERMITTIVITY ESTIMATION

Wave Propagation Effects:

- **Dispersion:** Phase velocity changes in plasma
- **Attenuation:** Signal strength reduction
- **Impedance Mismatch:** Antenna detuning

Frequency Comparison:

- **AIS (VHF ~162 MHz):** Near plasma cutoff → High susceptibility
- **S-band (~2 GHz):** Far above plasma cutoff → Low susceptibility
- **X-band (~8 GHz):** Minimal impact

CONTAMINATION MODELLING

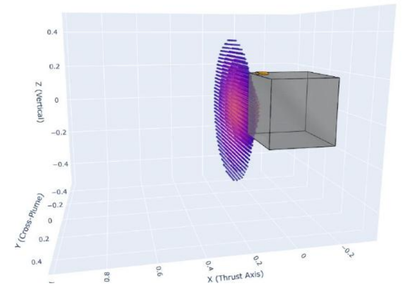
Mechanisms:

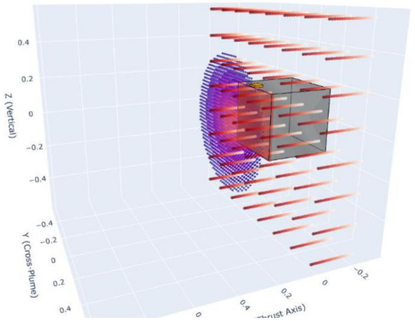
- Neutralized propellant deposition
- Charge-exchange ion accumulation

Effects:

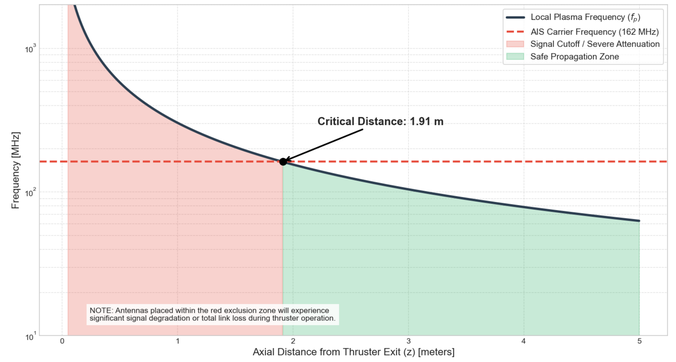
- Dielectric film formation on antenna surfaces
- Progressive antenna detuning over mission lifetime
- Irreversible performance degradation

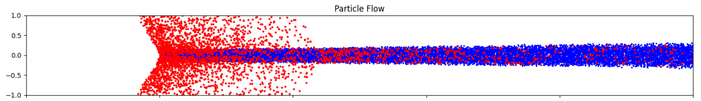
CHARGE EXCHANGE MODELLING





Practical AIS Antenna Placement: Plume Cutoff Analysis





Results

Parameter	VHF (AIS)	S-band	X-band
Carrier Frequency	~162 MHz	~2 GHz	~8 GHz
Plasma Frequency Ratio	$f_p \approx f$	$f_p \ll f$	$f_p \lll f$
Phase Distortion	**HIGH**	Low	Negligible
Signal Attenuation	**HIGH**	Low	Negligible
Impedance Mismatch	**HIGH**	Low	Negligible
Contamination Sensitivity	**HIGH**	Moderate	Low
Overall Susceptibility	**CRITICAL**	**MINOR**	**MINIMAL**

Key Findings & Design Implications

Key Results

- Plasma frequency **(~92 MHz)** approaches AIS frequency near thruster
- Attenuation: **26–28 dB** in dense plume regions
- Phase distortion**: ~35°** (GMSK signal degradation)
- Permittivity: **0.65–0.75** → impedance mismatch

RF System Sensitivity

- VHF (AIS): CRITICAL ($f_p \approx f$)
- S-band: LOW ($f_p << f$)
- X-band: MINIMAL ($f_p \lll f$)

Plume Effects

- Short-term: Plasma dispersion during firing (reversible)
- Long-term: CEX contamination → dielectric film → detuning (irreversible)

Impact on AIS

- Signal attenuation & phase errors
- Impedance mismatch and reduced efficiency

Design Mitigation

- Maximize antenna–thruster separation **(~2 m CRITICIAL distance)**
- Avoid AIS operation during thruster firing
- Apply contamination-resistant coatings
- Avoid plume backflow region

Mission Planning

- Schedule burns outside communication windows
- **Integrate plume–RF analysis in early design phase**

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